8.4 Metamorphic Rock **Name _____**

Metamorphic rock is rock that has been changed by p_____ and t_____ over long periods of time. I_____ and S_____ rocks can be changed into metamorphic rocks under these conditions. M_____ rocks can also be formed from other metamorphic rocks.

The changes can be p_____ or c_____, and include the formation of new m_____. Many gemstones such as r_____ and s_____ are found in metamorphic rock.

There are two types of metamorphism:

1. R_____ m_____ is metamorphism over a wide area.
2. C_____ m_____ occurs in particular local places on a much s_____ scale.

List five main steps involved in the formation of diamonds:

1. Parts of the crust _____
2. The crust sinks to depths of over 150km where high t_____ and p_____ change the rock.
3. The c_____ in the rock can c_____ to form diamonds.
4. Millions of years later the diamonds are carried to the surface by _____
5. The magma then _____ meaning that the diamonds are found in i_____ rocks even though they were not formed there.

What is foliation? _____

What does the amount of banding depend on? _____

List three metamorphic rocks that show banding _____

List two metamorphic rocks that do not show banding _____

Type of metamorphic rock	What it is used for
Quartzite and gneiss	B_____ work
Slate	Flat sheets used in r_____
Marble	Used by s_____ because it is easy to _____. It is also used in buildings for _____ and _____
Quartzite	Used for making g____ and stone tools.
Ruby, Sapphire, lapis lazuli, jade and garnet	G_____
Diamond	Used as gemstones in j_____. Used in drill _____.

Draw a diagram of the rock cycle that shows the three types of rocks and the agents that change them from one form to another.

Place the MM's and the jelly beans in the bottom of the patty pan.

Your teacher will pour some melted chocolate on top.

Now add the chocolate drops

Your teacher will pour some more melted chocolate on top.

Add the rice crispies.

Wrap some cling wrap around the bottom of the small beaker.

Use the base of the baker to squeeze the rice crispies down into the chocolate.

Eat your metamorphic rock.

1. In the making of metamorphic rock practical what did the hot chocolate represent?

2. What did the force of the beaker squeezing down on the chocolate and lolly mixture represent?